Limits of Resolution in Small Machine-Learning Benchmarks: A Noise Budget with a Quantum Case Study

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Abstract

Quantum machine learning (QML) benchmarks on small tabular datasets are usually read as rankings of architectures. We ask how large a difference such a benchmark can resolve. On a Western Mediterranean precipitation panel (four stations, 1938–2026; 1-month-ahead exceedance of a regional 60th percentile; 528 test station-months), one AUC has a standard error of about 0.024, 4–7 times the seed standard deviation of Random Forest, XGBoost and an LSTM, so additional seeds do not remove the uncertainty of a finite test sample. For differences of 15-seed means the minimum detectable difference is 0.031–0.044 AUC, and verdicts on smaller gaps depend on the procedure: for Random Forest–LSTM (+0.015) the year-block bootstrap interval excludes zero, whereas DeLong does not reject the null ($p = 0.19$). Known-truth simulations show DeLong calibrated with shared year effects but anticonservative with model-specific ones, and the year-block bootstrap moderately liberal. In a case study with matched fitting rows and 15 seeds per model, four hybrid quantum models scored 0.020–0.026 AUC below Random Forest and were not distinguished from a circuit-free classical control. No model improved on a month×station climatology: Random Forest matched its Brier score, the neural and hybrid models were worse, and within stations the climatology had the highest AUC. Alternative labels for the 23 % of incomplete test months preserved the signs of the differences but not their significance. We propose a reporting protocol for small QML benchmarks: a stated estimand, a noise budget with a minimum detectable difference, a climatological reference, matched fitting sets and a within-site evaluation.

Keywords: Quantum Machine Learning, Benchmarking, Statistical Power, Minimum Detectable Effect, Seed Variance, Precipitation Forecasting.

1 Introduction

Claims of quantum utility on classical tabular data frequently rest on a single training run per model, capacity-asymmetric classical baselines, and uncertainty estimates that cover only one source of error (Bowles et al., 2024). The machine-learning literature has long pointed out that a single run confounds initialization luck with architectural merit (Henderson et al., 2018; Bouthillier et al., 2021), and that tests computed on one fixed test set ignore the variance induced by the training procedure (Dietterich, 1998; Demšar, 2006). Variational circuits aggravate the issue: in the noisy intermediate-scale quantum (NISQ) era (Preskill, 2018), parameterized quantum circuits are sensitive to initialization and, at larger qubit counts, prone to barren plateaus (McClean et al., 2018), and they are usually combined with a classical backbone that compresses the input into a few qubits (Huang et al., 2021; Schuld & Killoran, 2022).

Throughout, "noise" means statistical variability (test-set sampling error and optimization-seed variance), not quantum hardware noise. The standard remedy is to repeat training with several seeds. This paper argues that, on small benchmarks, the remedy targets the smaller of the two error sources on the scale of an individual AUC: with a few hundred correlated test observations, the sampling error of each AUC of the classical models analyzed here is much larger than the variation caused by the seed, and the design can reliably detect only differences of a few hundredths of AUC. The qualification matters for QML: for the variational hybrid analyzed here, the seed is a substantial share of the variance of an AUC difference even after averaging 15 seeds, so seed and sampling error must both be reported. We also show that, at this scale, the conclusion about a small difference can depend on which inferential target and procedure are used, so these choices must be stated.

We demonstrate this on a deliberately modest but realistic case: a multi-decadal, multi-station Western Mediterranean precipitation panel, an area where heavy precipitation is linked to cut-off lows aloft (DANAs; Romero et al., 1999; Nieto et al., 2005) and where tree ensembles are strong baselines on engineered tabular features (Grinsztajn et al., 2022). The task is a 1-month-ahead probabilistic forecast of precipitation exceeding a pooled regional 60th percentile (P_{60} = 28.1 mm) from antecedent 12-month features. The panel is a testbed for evaluation methodology, not an operational early-warning system.

Applied climate- and weather-QML studies have addressed separate parts of the problem: opportunities for climate modeling (Schwabe et al., 2025a), parameter-count and initialization-matched comparisons for cloud cover (Schwabe et al., 2025b; Pastori et al., 2026), wind-speed forecasting (da Silva et al., 2026), quantum kernels for climate emulation (Bazgir & Zhang, 2024), synthetic time-series benchmarks (Fellner et al., 2026), variational data assimilation (Kumar et al., 2026), and parameter-matched neural comparisons (Marchisio et al., 2026). In the time-series and weather QML studies cited above, quantum models are reported as better than, or as having the potential to outperform, classical baselines on the basis of point estimates: Marchisio et al. (2026) report a 15.4 % lower MSE than a parameter-matched classical leaky-integrate-and-fire neuron on a 2,000-sample test set, with one value per model and no interval or test, and da Silva et al. (2026) use 10 runs per configuration and show their spread in violin plots, but compare the models through single error values with no interval or test, although their test sets span only 14 and 5 days. We do not claim that these effects are spurious; the point is that the evidence reported cannot separate them from sampling and seed noise. This paper complements these studies by asking how large a difference an evaluation design can resolve.

What the study shows. In a benchmark of 528 correlated observations, a protocol that separates test-sampling error, optimization seeds, temporal dependence and label validity changes qualitatively which conclusions can be drawn from AUC differences smaller than about 0.03; we propose and illustrate that protocol as a minimal evaluation standard for small QML benchmarks.

Contributions.

1. **A statistical framework** that separates two estimands, the difference of single-run AUCs and the difference of seed-averaged AUCs, and gives paired DeLong, year-block bootstrap and two-way (years and seeds) bootstrap inference for each, with the MDE as a design-sensitivity measure (Section 3).
2. **A noise budget** for Random Forest, XGBoost and an LSTM on 528 test station-months, comparing sampling error with seed variance (Section 4.1).
3. **Known-truth simulations** of two correlated scorers with shared and model-specific year effects, which check the type-I error, coverage and power of the procedures at this sample size (Section 4.2).
4. **An audit of label validity** for incomplete monthly totals, with the sensitivity of the labels and of the three classical comparisons to the label definition, by re-evaluating fixed predictors (Section 4.3).
5. **A QML case study and a reporting protocol**: an empirical benchmark in which four hybrid quantum models, a circuit-free classical control, a standalone LSTM and Random Forest, fitted on the same rows with 15 seeds each, are compared with the same paired procedures alongside a climatological reference (Sections 4.4 and 4.5), yielding the recommended reporting protocol collected in Section 5.2.

Further material (positive control, single-seed analyses, other hybrid architectures, ablations, operational metrics, circuit costs) is in the Supplementary Material.

2 Data, Task and Models

2.1 Dataset, Target and Split

The study domain is the Spanish Mediterranean coast (comprising Catalonia, the Valencian Community and the Region of Murcia), a transition zone between Mediterranean (*Csa*) and semi-arid (*BSk*/*BSh*) Köppen regimes subject to severe convective episodes (Romero et al., 1999; Nieto et al., 2005). Daily station records were compiled from historical series of the Spanish Meteorological Agency (AEMET) hosted by *datosclima.es* under the AEMET Open Data framework (daily series downloaded on 1 June 2026; station metadata from the AEMET climatological normals pages, accessed 28 October 2026). The stations are Alacant/Alicante (AEMET 8025; 38°22′21″ N, 0°29′39″ W; 81 m), Barcelona Aeropuerto (0076; 41°17′30″ N, 2°3′38″ E; 1 m), València (8416; 39°28′50″ N, 0°21′59″ W; 13 m) and San Javier Aeropuerto (7031; 37°47′20″ N, 0°48′12″ W; 4 m) (Table 1). Pressure is station-level pressure, not reduced to sea level, so its mean differs between stations by their altitude. The series span 1938–2026 at four stations (Barcelona, Valencia, Alicante and San Javier) and give 3,363 station-months; each station loses its first 12 months as the antecedent window (4×12 = 48), so the 3,363 station-months yield 3,315 prediction windows. Daily variables are aggregated to monthly resolution per station (precipitation sum,

wettest day, days above the daily 90th percentile, mean pressure, temperature, diurnal range, pressure variability, wind speed, maximum gust, reference evapotranspiration from FAO-56 Penman–Monteith with humidity estimated from the minimum temperature, radiation from sunshine duration and a constant wind speed $u_2 = 2\,\mathrm{m\,s^{-1}}$ (Allen et al., 1998), and sunshine duration). Months with fewer than 10 valid daily records are excluded, and monthly totals are sums over the available days; incomplete months therefore remain in the panel (Section 4.3). Inhomogeneities from relocations or instrument changes remain an accepted limitation (Alexandersson, 1986; Peterson et al., 1998).

Antecedent observations from months $i, \ldots, i+11$ (35 input features: 12 precipitation lags, 17 engineered physical features, 2 cyclic month terms and 4 station indicators) predict whether precipitation at month $i+12$ exceeds the pooled regional 60th percentile $P_{60} = 28.1$ mm, computed on the training partition only. The split is chronological: training 1938–1996 ($N = 2{,}273$ windows), validation 1997–2007 ($N = 514$) and test 2008–2026 ($N = 528$; positive rate 41.1 %, 217 positives). Station records are assembled by an inner join across five daily variable files, so San Javier (pressure ends 2010) and Valencia (2012) are under-represented in the test partition: 83.7 % of test samples come from Alicante and Barcelona (Table 1). Because test-period base rates differ strongly between stations (29–56 %; Table 1), part of the pooled AUC reflects base-rate differences between stations. Covariates are min–max scaled to $[0, \pi]$ with training-partition statistics only; re-computing the upstream monthly anomalies from pre-1997 data shifts the AUC of the standalone LSTM by 0.003 and, with the fitting rows held fixed, those of Random Forest and XGBoost by 0.003 or less (Section S2.4). Under train-only standardization the station-month anomalies (`iap`, `ia_racha`) are anchored to the pre-1997 climate, whereas the temperature anomaly `t_anomaly` and the derived contrast `choque_clausius` are z-scores relative to the trailing 12 months and are unaffected; other covariates, such as temperature range and evapotranspiration, are scaled with training statistics and may drift during accelerated warming, so the out-of-sample evaluation takes place under a covariate shift whose effect cannot be isolated with a single chronological split. All empirical conclusions concern the complete-variable panel retained by the processing pipeline, not the full set of station-months with precipitation observations.

Table 1: AEMET stations (climatological identifier, altitude), station-month windows by partition (training 1938–1996, validation 1997–2007, test 2008–2026) and test-set class prevalence for the regional exceedance threshold ($P_{60} = 28.1$ mm). Daily records span 1938-09 to 2026-05 (Alicante), 1961-01 to 2026-05 (Barcelona), 1951-01 to 2010-12 (San Javier) and 1938-02 to 2012-02 (Valencia).

Station	AEMET id (alt.)	Train	Val	Test	Total	Test positive rate
Alicante	8025 (81 m)	671	132	221	1,024	29.4% (65 / 221)
Barcelona	0076 (1 m)	378	132	221	731	50.2% (111 / 221)
San Javier	7031 (4 m)	535	132	36	703	36.1% (13 / 36)
Valencia	8416 (13 m)	689	118	50	857	56.0% (28 / 50)
Total		**2,273**	**514**	**528**	**3,315**	**41.1% (217 / 528)**

2.2 Models and Fitting Sets

We evaluate two reference baselines (a month×station climatology fitted on the training partition, and persistence), classical models (logistic regression, Random Forest with 300 trees (Breiman, 2001), gradient boosting, XGBoost with 500 estimators and depth 4, and a standalone bidirectional LSTM with 10,103 parameters), and hybrid quantum–classical models simulated noiselessly in PennyLane: `LSTM+QNN` (2,613

parameters, of which 24 quantum), `LSTM+PI-QNN`, `LSTM+QTL` (frozen LSTM, quantum transfer learning; Mari et al., 2020), `LbL-QNN` (Skolik et al., 2021) and a quantum-kernel SVM. The representative hybrid of this paper is `LSTM+QNN`: it compresses a 51-dimensional vector (16-dimensional LSTM state concatenated with the 35 engineered features) to 4 rotation angles before a 4-qubit circuit, which computes a truncated Fourier series of these angles (Schuld et al., 2021). The quantum-kernel SVM uses an unentangled product-state feature map whose kernel factorizes classically (Section S6.1). The controls of `LSTM+QNN` are three classical controllers on the same 4-dimensional bottleneck (linear, 20 parameters; LeakyReLU, 28; GELU, 180). Architectures, parameter counts and hyperparameters are given in Sections S2.7 and S3 and table S10.

The neural and hybrid models use the 2,273 training rows and the 514 validation rows for early stopping; the train-only Random Forest is fitted on the same 2,273 training rows. All models are scored on the same 528 test rows. The Random Forest of the classical train-only rerun and that of the matched QML core experiment of Section 4.5 are the same fits: their test probabilities agree to 3×10^{-16} for every seed, so the rerun Random Forest was fitted on the same 2,273 rows. The canonical-pipeline tree ensembles, in contrast, were fitted on training plus validation (2,787 rows). Refitting Random Forest in one software environment (scikit-learn 1.9.1, not bit-identical to the archived runs) gives 0.6875 (2,787 rows) and 0.6802 (2,273) in the canonical pipeline, and 0.6860 and 0.6831 in the train-only pipeline: the extra rows add 0.003–0.007 AUC, small relative to the test-sampling standard error of one AUC (about 0.024; Section 4.1).

Every gradient-optimized model and every tree ensemble is trained with 15 seeds; the seed index is nominal across frameworks. The primary metric is the ROC-AUC on the 528 test rows, complemented by the Brier score (Brier, 1950). The train-only pipeline, in which anomalies are standardized with pre-1997 data, is the primary one for Random Forest, XGBoost and the LSTM; `LSTM+QNN`, `LSTM+PI-QNN`, `LbL-QNN`, `LSTM+QTL` and the circuit-free control were re-run in the train-only pipeline on the same 2,273 rows (Section 4.5); the quantum-kernel SVM and the architecture ablations of the Supplement were fitted in the canonical pipeline only.

3 Statistical Framework

3.1 Estimands and Inference

For models A and B with test probabilities $p_{A,s}$, $p_{B,s}$ from seeds $s = 1, \ldots, S$ ($S = 15$), the main test statistic of this paper is the difference between seed-averaged test AUCs,

$$\hat{\Delta} = \frac{1}{S}\sum_{s} \widehat{\mathrm{AUC}}(p_{A,s}) - \frac{1}{S}\sum_{s} \widehat{\mathrm{AUC}}(p_{B,s}), \tag{1}$$

and, for single-run comparisons, the difference of two individual AUCs. Sampling inference targets the corresponding performance difference for the evaluation population, conditional on the observed training data and on the fitted seed-specific models. The DeLong variance additionally conditions on the observed test years and treats station-months as exchangeable; the year-block bootstrap and the simulations of Section 4.2 instead target the difference for new years, so a DeLong miscalibration in the simulations partly reflects this change of target. A bootstrap that also resamples seeds targets the seed-averaged difference

Table 2: Map of experiments: which pipeline, fitting rows, seeds and stored probabilities stand behind each result, and its inferential status.

Experiment	Pipeline	Fitting rows	Seeds	Stored test probabilities	Use (tables)
Core QML	train-only	2,273 (all models)	15	all seeds	primary (Tables 10 and 12 to 14)
Classical rerun	train-only (canonical branch for comparison)	2,273 (RF and LSTM identical to the core fits); canonical branch: trees 2,787	15	all seeds	primary (Tables 3 and 4)
Label sensitivity	saved predictions of the two rows above	none (no retraining)	15	all seeds	sensitivity (Tables 7 and 9)
Historical hybrids and ablations	canonical	trees 2,787 (RF also 2,273); neural 2,273	15 (AUCs)	none per seed; seed 0 regenerated (Section S1)	descriptive (Section S6)
Single runs	canonical	as above	1 (seed 42)	single run	descriptive (Section S6)

over the distribution of optimization seeds, again conditional on the training data. The mean of AUCs is linear in the DeLong structural components, so the paired sampling variance of $\hat{\Delta}$ is obtained by averaging the components over seeds; this treats the seeds as fixed.

3.2 Sampling Uncertainty

Test-set sampling error is estimated in three ways. (i) The paired DeLong estimator for correlated ROC curves (DeLong et al., 1988; Hanley & McNeil, 1982) treats station-months as independent and may therefore be optimistic for monthly data. (ii) A calendar-year block bootstrap resamples the 19 test years (2008–2026) with all stations of a year together ($B = 2{,}000$; Künsch, 1989; Obuchowski, 1997). Bootstrap confidence intervals are the 2.5th and 97.5th percentiles of the replicated statistic, and bootstrap standard errors are the sample standard deviations of the replicates; resamples that contain only one class are discarded. Where a bootstrap p-value is reported, it is $2\min\{\Pr(\hat{\Delta}^* \le 0), \Pr(\hat{\Delta}^* \ge 0)\}$ over the replicates $\hat{\Delta}^*$, bounded below by $1/B$. (iii) A two-way bootstrap resamples years and, within each model, seeds. With only 19 clusters the block bootstrap is itself imprecise and for some pairs returns a smaller standard error than DeLong; all procedures are reported, and none is assumed a priori to be the correct one for the observational panel. As a check on the block standard error, a delete-one-year jackknife recomputes $\hat{\Delta}$ with each of the $K = 19$ test years removed, $\mathrm{SE}_{\mathrm{jk}} = \{(K-1)/K \sum_k (\hat{\Delta}_{(k)} - \bar{\Delta}_{(\cdot)})^2\}^{1/2}$, with intervals $\hat{\Delta} \pm t_{0.975,18}\,\mathrm{SE}_{\mathrm{jk}}$ (Section 5.3).

3.3 Seed Variability and the Seed Share

For a model with seed standard deviation σ_{seed} and mean sampling standard error SE of one AUC we report $\sigma^2_{\mathrm{seed}}/(\sigma^2_{\mathrm{seed}} + \mathrm{SE}^2)$. This is a descriptive ratio of two conditional variances (seeds, on a fixed test set; test sampling, for a fixed trained model), not a decomposition of the total uncertainty of a training procedure; it compares the scale of the two sources. For a difference between models A and B, whose seeds are drawn independently, the seed variance is $\sigma^2_A + \sigma^2_B$ for one seed per model and $(\sigma^2_A + \sigma^2_B)/S$ for

the difference of S-seed means, and SE is the corresponding paired sampling standard error (for one seed per model, averaged over the $S \times S$ seed pairings).

3.4 Minimum Detectable Difference

For a paired difference with standard error SE_Δ, the minimum detectable difference at two-sided $\alpha = 0.05$ and power 0.80 is

$$\mathrm{MDE} = (z_{0.975} + z_{0.80})\,\mathrm{SE}_\Delta \approx 2.80\,\mathrm{SE}_\Delta. \tag{2}$$

Under a normal approximation with a locally constant standard error, the MDE describes the sensitivity of the design to a true difference: true differences below it are expected to be declared significant in fewer than 80 % of repeated experiments. It is not a threshold that an observed difference must exceed, and an observed gap between $1.96\,\mathrm{SE}_\Delta$ and $2.80\,\mathrm{SE}_\Delta$ can be nominally significant. We therefore report observed gaps with their intervals and use the MDE only to state which true differences the design would detect reliably.

3.5 Known-Truth Simulations and Scope of Claims

Two scorers with a known population AUC difference are simulated to check the type-I error, coverage and power of the procedures (design in Section 4.2). The noise budget, the paired comparisons, the simulation and the climatology comparison are the primary analyses of the paper, and the architecture ablations in the Supplement are secondary exploratory analyses. None was pre-registered. Holm correction is applied to the DeLong p-values within two families of three seed-mean comparisons (the classical models and the `LSTM+QNN` case); the comparisons of the other three hybrids (Table 14) were added after the main analysis and are secondary. Holm-adjusted p-values are the formal correction, and the first-step threshold 0.017 for three comparisons is quoted in table captions for reference. Bootstrap intervals are pointwise and not multiplicity-adjusted, and every nominal p-value outside these two families is hypothesis-generating. Seed-level tests (Wilcoxon, Friedman, paired t) compare seeds on one fixed test set, omit test sampling error, and are reported as statements about optimization variance.

4 Results

4.1 The Noise Budget

On the scale of one AUC, conditional test-sampling variability exceeds across-seed variability for all three classical models (Table 3): one AUC has a standard error of 0.023–0.027 under either estimator, whereas the seed standard deviation is 0.004–0.006, so the seed accounts for 2–7 % of the combined seed-plus-sampling variance (a descriptive comparison of scales, not an estimate of the share of total uncertainty that more seeds could remove). The hybrid quantum models have larger seed standard deviations. The 15-seed `LSTM+QNN` of Section 4.5 has a seed standard deviation of 0.0098 and a seed share of 15 % (12 % with the block standard error; last column of Table 3); for `LbL-QNN`, `LSTM+QTL` and `LSTM+PI-QNN`, fitted in the same experiment, the seed standard deviations are 0.0129, 0.0147 and 0.0170 and the seed shares 23, 27 and 34 % (block 18, 22 and 30 %). On the scale of one AUC the sampling term is the larger one for all the models analyzed, but for the hybrids the seed term is no longer small.

Table 3: Noise budget for one AUC, train-only pipeline, 15 seeds per model, 528 test rows. iid: DeLong; block: calendar-year block bootstrap (B = 2,000, 19 years). The seed share uses the DeLong standard error; with the block standard error it is 2, 4, 5 and 12 %. The classical columns are the rerun of Table 4; `LSTM+QNN` is the core experiment of Section 4.5; all models are fitted on the 2,273 training rows. Pairwise differences in Table 4 are computed from unrounded seed means.

Quantity	Random Forest	XGBoost	Standalone LSTM	LSTM+QNN
Mean AUC (15 seeds)	0.683	0.662	0.668	0.663
SE of one AUC, iid / block	0.023 / 0.026	0.024 / 0.025	0.024 / 0.027	0.024 / 0.027
SD across seeds	0.0035	0.0054	0.0063	0.0098
Seed share of combined variance	**2 %**	**5 %**	**7 %**	**15 %**

Table 4 and Figure 1 compare the 15-seed means (Δ = +0.015 for RF − LSTM; the Random Forest and LSTM fits are the same as in the core experiment of Section 4.5). The standard error of a paired difference is about half that of one AUC, and averaging the seeds lowers it further, but the DeLong MDE remains 0.031–0.044 (block estimates are 0.034, 0.017 and 0.043). For RF − XGBoost the difference is +0.021 (z = 1.93, p = 0.054) and both bootstrap intervals include zero: weak, borderline evidence. For RF − LSTM both bootstrap intervals exclude zero by a small margin, but DeLong does not (p = 0.189), and the block standard error (0.006) is about half of the DeLong one (0.011). With 19 clusters the block estimate is itself imprecise, and its MDE (0.017) is substantially smaller than the iid one; given the limited number of clusters and the uncertainty about the dependence structure, neither estimate is treated as definitive. We therefore describe this comparison as dependent on the inferential procedure and not as established or refuted. XGBoost against the LSTM shows no evidence of a difference, and none of the three DeLong comparisons rejects the null after Holm correction over the three pairs (smallest p = 0.054 against a first-step threshold of 0.017). The bootstrap intervals in Table 4 are pointwise 95 % intervals, are not multiplicity-adjusted, and no bootstrap p-values were computed for these pairs.

Pairing removes most of the sampling error shared by two models but none of the seed variance, so the seed share is larger for a difference than for one AUC. For the classical pairs, the seed accounts for 17–22 % of the combined variance of a single-seed difference (17–35 % with the block standard error, the upper value for RF − LSTM) and for 2–3 % (2–9 %) of that of a difference of 15-seed means. For the hybrid of Section 4.5 the seed is not a minor term: in its differences with Random Forest, the linear control and the standalone LSTM, the seed share is 32–47 % (block 45–47 %) for one seed per model and still 6–27 % (block 13–33 %) for the 15-seed means, the largest values arising against the standalone LSTM; for the other three hybrids the corresponding shares for the 15-seed means are 8–13 % against Random Forest (block 16–32 %) and 25–47 % against the two neural comparators (block 17–53 %). The DeLong and year-block intervals condition on the fitted seeds and omit this component; the years×seeds bootstrap includes it, which is why its intervals in Table 13 are wider than the year-block ones.

Table 4: Paired comparison of the 15-seed means (train-only pipeline): $\Delta = \overline{\mathrm{AUC}}_A - \overline{\mathrm{AUC}}_B$ with the paired DeLong standard error on seed-averaged components, two-sided p-value, 95 % intervals from the calendar-year block bootstrap (19 years, B = 2,000) and from a two-way bootstrap that resamples years and seeds, and the MDE (2.80× SE) under DeLong and the block bootstrap.

Pair	Δ	SE (DeLong)	z (p)	Block 95 % CI (SE)	Two-way 95 % CI	MDE iid / block
RF − XGBoost	+0.021	0.0110	1.93 (0.054)	[−0.002, +0.045] (0.0122)	[−0.003, +0.046]	0.031 / 0.034
RF − LSTM	+0.015	0.0112	1.31 (0.189)	[+0.002, +0.025] (0.0061)	[+0.001, +0.027]	0.031 / 0.017
XGBoost − LSTM	−0.007	0.0159	−0.42 (0.677)	[−0.039, +0.022] (0.0153)	[−0.037, +0.023]	0.044 / 0.043

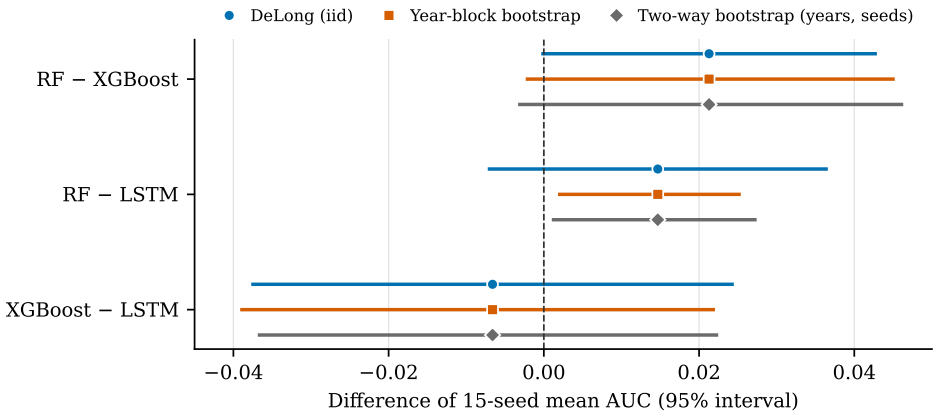


Figure 1: Differences of 15-seed mean AUC with 95 % intervals from the three procedures (train-only pipeline, 528 test rows). The procedures agree for RF − XGBoost and XGBoost − LSTM and disagree on whether the RF − LSTM interval excludes zero.

Single-seed comparisons are less stable. The paired DeLong standard error of two single runs is 0.013–0.018 (MDE 0.036–0.052), and in the train-only pipeline RF > XGBoost is declared significant in 27 % of the 15 × 15 seed pairings (21 % under the block bootstrap), RF > LSTM in 3 % (24 %) and XGBoost versus LSTM in 0 % (Table S1). A verdict about a single pair of runs therefore depends largely on which seeds happen to be compared (Section S1.1).

4.2 Simulation with Known Truth

The positive control of the Supplement perturbs observed probabilities, so it cannot establish the coverage or power of the procedures over new samples. We therefore simulate two scorers whose population AUC difference is known by construction or approximated on large reference samples (four independent draws of 400,000 rows; Monte Carlo standard error of the population difference at most 0.0002). Labels have prevalence 0.41 with year-level effects on the logit, and the 528 rows are split among 19 years as equally as possible or, in one scenario, unequally (multinomial with Dirichlet(3) shares). The scorers are binormal with noise correlation $\rho = 0.9$, which reproduces the paired DeLong standard error of the classical comparisons (0.011), and their design AUCs without year effects are 0.65 and 0.65 + Δ. Year effects enter as a prevalence shift (τ_π), a shift shared by both scorers (τ_s) and, in scenarios C–E, a shift specific to each scorer and independent between them (τ_m). Scenario B has $\tau_\pi = \tau_s = 0.3$, C adds $\tau_m = 0.3$, D is C with unequal year sizes, and E has all three equal to 0.6. The target is the pooled population difference for new years drawn with equal weight (details in Section A). When the design difference is 0 the scorers are exchangeable and the population difference is exactly 0. Each scenario has 500 replications; the block bootstrap uses $B = 300$ (Section A). Rejection rates have a Monte Carlo standard error of about 0.01 near 0.05.

Table 5 and Figure 2 give three results. First, with year effects shared by the two scorers (scenario B) DeLong is calibrated (type-I error 0.040, 20 of 500; coverage 0.960; standard deviation of the estimate 0.0108 against a mean standard error of 0.0111), whereas the block bootstrap rejects the exact null in 7.4 % of replications (Monte Carlo standard error 1.2 points) with coverage 0.926. Second, with scorer-specific year effects the DeLong standard error underestimates the standard deviation of the estimate

Table 5: Simulation with known truth ($N = 528$, 19 years, prevalence 0.41, $\rho = 0.9$, 500 replications). τ_π, τ_s and τ_m: standard deviations of the year effects on the prevalence logit, on a shift shared by both scorers and on a shift specific to each scorer. Type-I error and coverage are for an exactly null difference (DeLong, block bootstrap); SD($\hat{\Delta}$) is the Monte Carlo standard deviation of the estimate, to be compared with the mean standard error (DeLong / block). Last column: power at the largest simulated difference (population difference in brackets). All rows are in the Supplement.

Scenario	Type-I DeLong	Type-I block	Cover. DeLong	Cover. block	SD($\hat{\Delta}$)	Mean SE	Power (DeLong / block)
B: shared year effects ($\tau_\pi = \tau_s = 0.3$)	0.040	0.074	0.960	0.926	0.0108	0.0111 / 0.0108	0.74 / 0.78 (0.029)
C: + scorer-specific year effects ($\tau_m = 0.3$)	0.068	0.076	0.932	0.924	0.0154	0.0144 / 0.0153	0.50 / 0.47 (0.028)
D: C with unequal year sizes	0.096	0.084	0.904	0.916	0.0168	0.0144 / 0.0155	0.51 / 0.48 (0.028)
E: all $\tau = 0.6$	0.220	0.074	0.780	0.926	0.0290	0.0185 / 0.0273	0.35 / 0.21 (0.023)

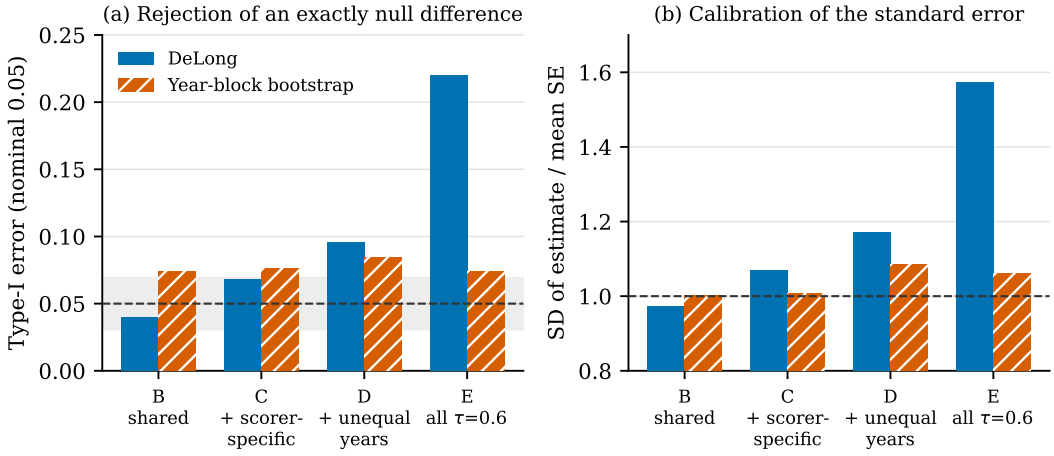


Figure 2: Simulation scenarios B–E (Table 5). (a) Rejection rate of an exactly null difference; the band is the 95 % Monte Carlo interval around the nominal 0.05 for 500 replications. (b) Ratio of the Monte Carlo standard deviation of the estimate to the mean standard error; values above 1 mean that the standard error is too small.

(ratio 1.07 in C, 1.17 in D with unequal years, 1.57 in E) and its type-I error rises to 0.068, 0.096 and 0.220 with coverage 0.932, 0.904 and 0.780; the block standard error tracks the standard deviation (ratios 1.01, 1.08 and 1.06), and its type-I error stays at 0.076, 0.084 and 0.074 with coverage 0.924, 0.916 and 0.926. Neither procedure is therefore uniformly better calibrated: DeLong fails when year effects are scorer-specific, and the block bootstrap with 19 clusters is less sensitive to the scenario but liberal (type-I error 0.074–0.084, coverage 0.916–0.926), so nominal coverage was not achieved uniformly in these scenarios. Additionally, in scenarios C and E at intermediate differences ($\Delta_{\text{true}} \approx 0.011$–$0.014$) the empirical mean of the finite-sample estimates showed a slight positive displacement of 0.0025–0.0040 above the reference (about 3 Monte Carlo standard errors), contributing to the undercoverage. Third, power follows the MDE rule where DeLong is calibrated: in scenario B (standard error 0.011, MDE 0.031) a true difference of 0.029 is detected in 74 % of replications (block 78 %) and 0.0145 in 32 % (35 %); in scenarios C–E the larger standard deviation of the estimate (0.015–0.029) raises the MDE accordingly. The mechanism is Gaussian and binormal with equal-weight years and no spatial structure or optimization noise; it checks the procedures at this sample size and does not reproduce the dependence of the real panel.

Panel-matched check. To assess whether the stylized scenarios misrepresent the panel, we repeated the simulation using the observed year sizes (19 years, 10–48 rows per year), the observed test prevalence (0.411), the AUCs of the seed-averaged RF and LSTM predictions (0.684 and 0.673; observed ensemble gap 0.0115, a third estimand distinct from the +0.015 difference of seed-mean AUCs of Section 3.1), and $\rho = 0.89$, estimated from their within-class normal scores. The year-level variance of both prevalence and scores was not distinguishable from sampling noise: the method-of-moments estimates were negative and were therefore truncated to zero. We therefore varied a common year-effect scale τ from 0 to 0.3 (prevalence-logit and shared-score shifts with standard deviation τ, scorer-specific shifts with $\tau/2$), with 500 replications per value. Across this grid, the type-I error of the paired DeLong test was 0.034–0.056 (coverage 0.944–0.966), whereas that of the calendar-year block bootstrap was 0.054–0.092 (coverage 0.908–0.946). The mean DeLong and block standard errors, 0.0117–0.0123 and 0.0113–0.0125, were close to the empirical standard deviation of $\hat{\Delta}$, 0.0109–0.0132. These standard errors imply an approximate 80 %-power MDE of 0.033. At the observed ensemble gap of 0.0115 the estimated power was only 0.14–0.19 for DeLong and 0.17–0.22 for the block bootstrap; a difference of that size would thus be detected in fewer than 25 % of replications at this sample size. As in the other scenarios, this check cannot identify the year effects present in the real panel and omits spatial structure and optimization noise.

Conditional perturbation (positive control). In Sections S1.2 and S1.3, the test probabilities of Random Forest are degraded by calibrated logit noise until the AUC falls by a known amount, and the detection rate of the undegraded model is recorded. The 80 % detection rate is reached at an imposed AUC reduction of about 0.04 and 17–30 % of runs detect a reduction of 0.02, consistent with the DeLong MDE of single-seed pairings. This is a perturbation of observed probabilities, not a simulation from a population, and it does not establish coverage or power over new samples.

4.3 Label Validity under Incomplete Months

Monthly precipitation is the sum of the available daily values, and months with at least 10 valid days are retained. Of the 3,363 retained station-months, 327 (9.7 %) are incomplete (fewer valid days than days

in the month), and the share is much higher in the test partition (123 of 528, 23.3 %; 7.5 % in training) and in Barcelona (217 of 743 months, 29.2 %). An incomplete month can only under-count precipitation, so some incomplete months labeled negative may in fact exceed the threshold. Table 6 recomputes P_{60} and the labels under stricter criteria. Requiring 15, 20 or 25 valid days changes 0, 3 and 8 labels among the common months (0, 0 and 2 in test); keeping complete months only changes 17 labels (2 in test) but removes 123 of the 528 test rows, so this check mixes label changes with a change in test composition. A proportional-coverage scenario, which scales each monthly sum by days in month over valid days, assumes that missing days rain like observed days; it is a sensitivity scenario and neither a validated correction nor a bound. It moves the threshold from 28.0 to 28.4 mm, changes 32 labels (14 in test) and raises the test prevalence to 0.430; at the project threshold, 18 incomplete months (12 in test, 2.3 % of test rows) change label between the plain and the corrected sum.

Table 6: Sensitivity of the monthly labels to the coverage criterion. Flips are labels that differ from the project labels among the common months (in brackets: test partition). Prevalence and n refer to the test partition.

Criterion	Months kept	P_{60}	Flips (test)	Test prevalence	Test n
Project: ≥ 10 days, plain sum	3,363	28.0	–	0.411	528
≥ 15 days, plain sum	3,354	28.1	0 (0)	0.413	523
≥ 20 days, plain sum	3,336	28.1	3 (0)	0.416	512
≥ 25 days, plain sum	3,284	28.3	8 (2)	0.417	489
Complete months only	3,036	27.3	17 (2)	0.407	405
≥ 10 days, proportional-coverage scenario	3,363	28.4	32 (14)	0.430	528

The audit recomputes P_{60} from the training station-months of the monthly table (28.0 mm) and not from the windowed dataset (28.1 mm); the labels obtained with 28.0 mm agree with the dataset labels on all 528 test rows, so this difference changes no test label.

Observed rain discarded by the join. For the 3,363 station-months of the panel, we compared the precipitation summed over the days retained by the inner join across the five daily variable files with the sum over all days with observed precipitation in the precipitation-only files (the latter is not a complete monthly total when days are also missing there). Of these station-months, 325 lose at least one day (1,299 days in total) and 121 (3.6 %) lose observed rain, 1,776.6 mm in total; among the affected months the median loss is 6.8 mm, the 95th percentile 55.4 mm and the maximum 125.4 mm. The loss is concentrated in Barcelona (83 of the 121 months) and in the test partition (56 months, 666.2 mm; Valencia's 50 test months are unaffected). At the fixed project threshold 20 labels change (9 in test, 11 in training, none in validation); with the threshold recomputed from the unjoined training sums (28.6 mm instead of 28.0 mm) 41 labels change (12 in test). A further 897 station-months with at least 10 days of observed precipitation do not enter the panel because fewer than 10 days survive the join; this is a selection of months and not a change of labels, and its effect was not examined.

Whether the model comparisons change is examined with fixed predictors and without retraining (scenarios S0–S7; Tables 7 to 9). Three kinds of re-evaluation are reported: evaluation on a subset of rows with the project labels (S1–S3), evaluation of all rows against labels from the proportional-coverage sum (S4, S5), and against labels from the unjoined sum (S6, S7). Retraining under another label definition or pipeline was not done. The predictions are those of the train-only pipeline, 15 seeds per model; the hybrid case includes the 15-seed `LSTM+QNN` of the core experiment and its comparators (Table 9); the other three

Table 7: Re-evaluation of the saved test predictions (15-seed means, no retraining) under alternative label definitions: sample, labels and scores of the classical models. S4–S7 are hypothetical scenarios that evaluate predictors trained on the project labels against alternative labels. Pos.: positive test labels. Flips: test labels that differ from the project labels. Differences between models are in Tables 8 and 9.

Scenario	n	Pos.	Prev.	Flips	AUC RF / XGB / LSTM	Brier RF / XGB / LSTM
S0 project labels, all rows	528	217	0.411	0	0.683 / 0.662 / 0.668	0.224 / 0.246 / 0.239
S1 complete target months	405	163	0.402	0	0.695 / 0.681 / 0.689	0.219 / 0.234 / 0.228
S2 ≥ 25 valid days	489	206	0.421	0	0.695 / 0.678 / 0.681	0.221 / 0.238 / 0.233
S3 ambiguous labels dropped	516	217	0.421	0	0.688 / 0.667 / 0.676	0.222 / 0.243 / 0.236
S4 coverage-corrected, same threshold	528	229	0.434	12	0.688 / 0.667 / 0.678	0.223 / 0.243 / 0.234
S5 coverage-corrected, new threshold	528	227	0.430	14	0.690 / 0.670 / 0.680	0.222 / 0.242 / 0.233
S6 unjoined sum, same threshold	528	226	0.428	9	0.686 / 0.660 / 0.679	0.223 / 0.246 / 0.234
S7 unjoined sum, new threshold	528	221	0.419	12	0.684 / 0.661 / 0.679	0.223 / 0.246 / 0.235

hybrids are not tabulated.

For the classical models the sign of every paired difference is the same in all eight scenarios (RF − XGBoost +0.014 to +0.026; RF − LSTM +0.005 to +0.015; XGBoost − LSTM −0.003 to −0.019), but their nominal significance is not. RF − XGBoost has DeLong p between 0.019 and 0.285, below 0.05 only in S6 and S7 (unjoined-sum labels), and a year-block interval that excludes zero only in S6 ($p = 0.019$, not below the Holm first-step threshold of 0.017 for three pairs). RF − LSTM has p between 0.189 and 0.622, never below 0.05, and, among the eight scenarios, a year-block interval that excludes zero only under the original labels, S0; with unjoined-sum labels its difference falls to +0.005–+0.007. The year-block intervals come from one bootstrap run ($B = 2{,}000$); a second run with a different random-number stream moved the endpoints of the classical pairs by up to 0.0027, and the lower limit of the S0 interval of RF − LSTM is +0.0006 here and +0.0018 in Table 4. Both are positive, but the margin by which this interval excludes zero is smaller than the run-to-run variation of its endpoints, so the exclusion is narrow and should not be interpreted as robust evidence. The comparisons that are borderline in the main analysis thus move in opposite directions under plausible label definitions, and none reaches nominal significance in a way that is stable across them. Restricting to complete months raises the AUC of the three models by 0.01–0.02 but reduces the test size to 405 rows. The Brier scores keep the same order (Random Forest lowest) in all scenarios.

For the hybrid case (Table 9), the RF − `LSTM+QNN` difference keeps its sign (+0.011 to +0.020) but not its interval-based verdict: the DeLong p ranges over 0.074–0.367 and is never below 0.05; the year-block interval excludes zero in S0 and S2–S5 and includes it in S1 (complete months) and in S6 and S7 (unjoined-sum labels). The Linear-4D control scores below the hybrid by 0.005–0.009 in every scenario (year-block intervals from −0.032 to +0.011), and the standalone LSTM above it by +0.003 to +0.007, with intervals that include zero in all scenarios for both pairs.

4.4 Reference Baselines: Climatology and Probabilistic Skill

A month×station climatology fitted on the training partition reaches AUC 0.680 (Brier score 0.2224; Brier skill score +0.082 relative to the training base rate of 0.4017, that is, to the Brier score 0.2422 of the constant forecast 0.4017 evaluated on the test rows, whose prevalence is 0.411); persistence is weaker (lag-1 precipitation AUC 0.589). Random Forest exceeds the climatology by +0.003 AUC (range −0.002 to +0.011 across seeds) and no seed is significantly better under either test; XGBoost and the LSTM score

Table 8: Paired differences of the classical models under alternative label definitions (Δ, seed-mean AUC difference) with year-block 95 % interval and DeLong p (seed-averaged components, unadjusted). The year-block intervals come from one bootstrap run (B = 2,000) of the same script; a second run with a different random-number stream moved interval endpoints of these pairs by up to 0.0027, and the S0 interval of RF − LSTM differs from Table 4 (lower limit +0.0006 against +0.0018).

Scenario	RF − XGB: Δ [block CI]; p	RF − LSTM: Δ [block CI]; p
S0 project labels, all rows	+0.021 [−0.002, +0.045]; 0.054	+0.015 [+0.001, +0.026]; 0.189
S1 complete target months	+0.014 [−0.012, +0.041]; 0.285	+0.006 [−0.010, +0.024]; 0.617
S2 ≥ 25 valid days	+0.018 [−0.005, +0.041]; 0.112	+0.015 [−0.001, +0.029]; 0.191
S3 ambiguous labels dropped	+0.021 [−0.003, +0.044]; 0.054	+0.012 [−0.0002, +0.023]; 0.265
S4 coverage-corrected, same threshold	+0.021 [−0.003, +0.043]; 0.052	+0.010 [−0.003, +0.023]; 0.358
S5 coverage-corrected, new threshold	+0.020 [−0.003, +0.043]; 0.064	+0.010 [−0.003, +0.022]; 0.375
S6 unjoined sum, same threshold	+0.026 [+0.002, +0.049]; 0.019	+0.007 [−0.009, +0.021]; 0.541
S7 unjoined sum, new threshold	+0.023 [−0.001, +0.048]; 0.035	+0.005 [−0.010, +0.020]; 0.622

Table 9: The hybrid case (Section 4.5) under the same label definitions: AUC of Random Forest (2,273 rows), standalone LSTM, Linear-4D control and `LSTM+QNN`, and paired differences with `LSTM+QNN` (Δ [year-block 95 % CI]; unadjusted DeLong p). Holm first-step threshold for these three pairs: 0.017.

Scenario	AUC RF / LSTM / Linear-4D / QNN	RF − QNN	Linear-4D − QNN	LSTM − QNN
S0 project labels, all rows	0.683 / 0.668 / 0.654 / 0.663	+0.020 [+0.007, +0.031]; 0.074	−0.009 [−0.028, +0.006]; 0.223	+0.005 [−0.003, +0.013]; 0.334
S1 complete target months	0.695 / 0.689 / 0.675 / 0.684	+0.011 [−0.006, +0.028]; 0.367	−0.009 [−0.032, +0.010]; 0.317	+0.004 [−0.003, +0.012]; 0.417
S2 ≥ 25 valid days	0.695 / 0.681 / 0.669 / 0.678	+0.017 [+0.001, +0.034]; 0.112	−0.009 [−0.028, +0.006]; 0.250	+0.003 [−0.005, +0.012]; 0.605
S3 ambiguous labels dropped	0.688 / 0.676 / 0.662 / 0.670	+0.018 [+0.006, +0.029]; 0.095	−0.009 [−0.026, +0.008]; 0.273	+0.006 [−0.002, +0.015]; 0.251
S4 coverage-corrected, same threshold	0.688 / 0.678 / 0.664 / 0.671	+0.017 [+0.004, +0.029]; 0.121	−0.007 [−0.028, +0.011]; 0.346	+0.007 [−0.003, +0.016]; 0.176
S5 coverage-corrected, new threshold	0.690 / 0.680 / 0.666 / 0.674	+0.017 [+0.005, +0.029]; 0.128	−0.007 [−0.027, +0.011]; 0.343	+0.007 [−0.002, +0.016]; 0.174
S6 unjoined sum, same threshold	0.686 / 0.679 / 0.668 / 0.674	+0.012 [−0.004, +0.027]; 0.258	−0.005 [−0.023, +0.011]; 0.479	+0.006 [−0.002, +0.014]; 0.266
S7 unjoined sum, new threshold	0.684 / 0.679 / 0.667 / 0.673	+0.012 [−0.004, +0.027]; 0.292	−0.006 [−0.024, +0.011]; 0.468	+0.006 [−0.002, +0.014]; 0.232

below the climatology on point estimates (−0.018 and −0.011), also without significance. Two further references require no retraining. A station-only base rate fitted on the training partition reaches AUC 0.599 (Brier score 0.2392), about half of the climatology's excess over 0.5; adding the month raises the AUC by 0.080 (DeLong $p < 0.001$). The same month×station climatology fitted on the training and validation years (1938–2007, 2,787 rows), a reference that uses more years than the 2,273 rows on which Random Forest is fitted, reaches AUC 0.686 (Brier score 0.2186). Against this training-plus-validation climatology the 15-seed Random Forest mean differs by −0.003 (DeLong $p = 0.83$; year-block 95 % CI [−0.025, +0.016]): the advantage of Random Forest over the training-only climatology is smaller than what the climatology itself gains from the validation rows (+0.006, $p = 0.27$). On proper scoring (Table 10; 15-seed train-only predictions, no retraining), no model improves on the climatological Brier score on average. Random Forest matches it (mean Brier score 0.2238 against 0.2224; Brier skill score −0.006, from −0.013 to +0.001 across seeds, with one seed of 15 marginally below the climatology; year-block 95 % interval of the difference [−0.006, +0.009]). XGBoost, the standalone LSTM, the Linear-4D control and the four variational hybrids, all trained with positive-class weighting, score 0.017–0.023 worse than the climatology (skill −0.075 to −0.104; every year-block interval excludes zero) and have higher expected calibration errors (ECE 0.120–0.137, against 0.075 for Random Forest; whether they are over- or under-confident was not analyzed). The canonical single-run evaluation, including logistic regression, gradient boosting and the quantum-kernel SVM, gives the same picture (Sections S6.2 and S6.3).

Within-station discrimination. Because the threshold is pooled across stations whose base rates differ, the pooled AUC partly rewards telling stations apart. The within-station AUC counts only positive–

Table 10: Probabilistic skill against the month×station climatology (train-only pipeline, 15 seeds per model, 528 test rows; archived predictions, no retraining; script `exp21_brier_trainonly.py`). Brier score: 15-seed mean (SD). $\mathrm{BSS}_{\mathrm{clim}} = 1 - \mathrm{BS}/0.2224$: seed mean [minimum, maximum]. ΔBS: mean Brier score minus that of the climatology, with year-block 95 % interval (B = 2,000). ECE: 10-bin expected calibration error, seed mean. The canonical single-run version of this table, which also includes logistic regression, gradient boosting and the quantum-kernel SVM, is Table S16.

Model	AUC	Brier score	$\mathrm{BSS}_{\mathrm{clim}}$	ΔBS [year-block CI]	ECE
Month×Station Climatology	0.680	0.2224	0	–	0.089
Random Forest	0.683	0.2238 (0.0009)	−0.006 [−0.013, +0.001]	+0.001 [−0.006, +0.009]	0.075
XGBoost	0.662	0.2456 (0.0025)	−0.104 [−0.127, −0.085]	+0.023 [+0.009, +0.038]	0.125
Standalone LSTM	0.668	0.2395 (0.0046)	−0.077 [−0.109, −0.033]	+0.017 [+0.007, +0.027]	0.137
Linear-4D control	0.654	0.2429 (0.0042)	−0.092 [−0.117, −0.051]	+0.021 [+0.010, +0.032]	0.131
LSTM+QNN	0.663	0.2421 (0.0061)	−0.088 [−0.126, −0.053]	+0.020 [+0.011, +0.028]	0.131
LbL-QNN	0.663	0.2393 (0.0044)	−0.076 [−0.111, −0.052]	+0.017 [+0.008, +0.025]	0.128
LSTM+PI-QNN	0.660	0.2391 (0.0067)	−0.075 [−0.119, −0.025]	+0.017 [+0.007, +0.026]	0.125
LSTM+QTL	0.657	0.2400 (0.0069)	−0.079 [−0.128, −0.030]	+0.018 [+0.008, +0.027]	0.120

negative pairs of months from the same station (the per-station AUCs averaged with weights $n_{\mathrm{pos}} n_{\mathrm{neg}}$); a station-only predictor scores exactly 0.5 on it (Table 11; `exp23_within_station_auc.py`). On this scale every learned model loses 0.020–0.035, the climatology only 0.010, and the month×station climatology has the highest value (0.670). Random Forest is not distinguished from it (−0.010, year-block 95 % CI [−0.034, +0.009]), whereas `LSTM+QNN` is below it (−0.031, [−0.060, −0.005]). The Random Forest − `LSTM+QNN` difference keeps its size (+0.022) but its year-block interval now reaches zero ([−0.0003, +0.039]). Within a station, therefore, none of the learned models adds discrimination beyond the seasonal cycle encoded by the climatology.

Table 11: Pooled and within-station AUC (train-only pipeline, 15-seed means, 528 test rows; archived predictions, no retraining). Within-station AUC: positive–negative pairs of the same station only. SD: across seeds. Per-station AUC in the last four columns (test n: 221, 221, 36, 50).

Model	Pooled AUC	Within-station AUC (SD)	Alicante	Barcelona	San Javier	Valencia
Month×Station Climatology	0.680	0.670 (–)	0.681	0.663	0.555	0.692
Random Forest	0.683	0.660 (0.005)	0.703	0.628	0.555	0.646
XGBoost	0.662	0.627 (0.007)	0.675	0.586	0.558	0.667
Standalone LSTM	0.668	0.648 (0.012)	0.674	0.628	0.585	0.661
Linear-4D control	0.654	0.620 (0.014)	0.644	0.598	0.619	0.652
LSTM+QNN	0.663	0.639 (0.015)	0.681	0.604	0.590	0.642
LbL-QNN	0.663	0.638 (0.018)	0.677	0.606	0.570	0.646
LSTM+PI-QNN	0.660	0.634 (0.026)	0.674	0.601	0.567	0.659
LSTM+QTL	0.657	0.631 (0.025)	0.657	0.611	0.565	0.654

4.5 Case Study: A Hybrid Quantum Model under the Noise Budget

The core experiment fits, in the train-only pipeline and on the same 2,273 training rows (the validation rows are used only for early stopping), Random Forest, the standalone LSTM, a classical control with the architecture of the hybrid in which the 4-qubit circuit is replaced by a linear $4 \to 4$ map (20 parameters), the hybrid `LSTM+QNN` (24 quantum parameters) and the three further hybrids `LSTM+PI-QNN`, `LbL-QNN` and `LSTM+QTL`, with 15 seeds each and the test probabilities of every seed stored. Table 12 gives the seed means. The models share the 2,273 fitting rows, the validation rows for early stopping (gradient-trained models), the 528 test rows and the 15 seed indices, which are nominal across frameworks; sharing rows

does not imply equal capacity, tuning budget or input representation: the hyperparameters, feature sets, stopping rules, seeds and output files of each model are given in Table S3 of the Supplement and defined in `qml_core_trainonly.py` and `exp17_hybrids_multiseed.py`. The dataset is the one of the classical rerun: the Random Forest and standalone LSTM fits reproduce those of Table 4 to within 10^{-11} in every test probability. Random Forest has the highest mean, every hybrid lies below it and below the standalone LSTM, and the hybrids have seed standard deviations 1.5–2.7 times that of the standalone LSTM. The paired comparisons of the seed means use the procedures of Section 4.1 (Tables 13 and 14).

Table 12: 15-seed test AUC (train-only pipeline, same 2,273 fitting rows; mean and sample SD), sorted by mean. The last column is the difference to `LSTM+QNN`; it is descriptive, not a test. The frozen extractor of `LSTM+QTL` is the standalone LSTM of the same seed (identical test probabilities). Climatology: AUC 0.6796.

Model	Mean AUC	SD	Gap to LSTM+QNN
Random Forest (2,273 rows)	0.6829	0.0035	+0.020
Standalone LSTM	0.6682	0.0063	+0.005
LSTM+QNN (24 quantum parameters)	0.6634	0.0098	0
LbL-QNN	0.6632	0.0129	−0.000
LSTM+PI-QNN	0.6601	0.0170	−0.003
LSTM+QTL	0.6565	0.0147	−0.007
Linear-4D classical control	0.6539	0.0078	−0.009

Table 13: Paired comparisons of the 15-seed means with `LSTM+QNN` (train-only pipeline, 528 test rows, 19 test years; same procedures and notation as Table 4). DeLong p-values are unadjusted; the Holm first-step threshold for these three pairs is 0.017. The MDEs are specific to each pair.

Difference	$\hat{\Delta}$	SE (DeLong)	z (p)	Year-block 95 % CI (SE)	Years×seeds 95 % CI	MDE DeLong / block
RF − LSTM+QNN	+0.020	0.0109	1.79 (0.074)	[+0.007, +0.031] (0.0062)	[+0.006, +0.034]	0.031 / 0.017
Linear-4D − LSTM+QNN	−0.009	0.0077	−1.22 (0.223)	[−0.026, +0.006] (0.0081)	[−0.030, +0.008]	0.022 / 0.023
LSTM − LSTM+QNN	+0.005	0.0050	0.97 (0.334)	[−0.003, +0.013] (0.0042)	[−0.007, +0.018]	0.014 / 0.012

Table 14: Paired comparisons of the 15-seed means with the three further hybrids (same experiment, procedures and notation as Table 13). For `LSTM+QTL` the comparator is its own frozen extractor, which is the standalone LSTM of the same seed. Secondary analysis: DeLong p-values are unadjusted and the intervals pointwise.

Difference	$\hat{\Delta}$	SE (DeLong)	z (p)	Year-block 95 % CI (SE)	Years×seeds 95 % CI	MDE DeLong / block
RF − LSTM+PI-QNN	+0.023	0.0116	1.96 (0.050)	[+0.010, +0.036] (0.0066)	[+0.005, +0.039]	0.033 / 0.019
RF − LbL-QNN	+0.020	0.0116	1.70 (0.089)	[+0.004, +0.035] (0.0080)	[+0.002, +0.038]	0.033 / 0.022
RF − LSTM+QTL	+0.026	0.0118	2.25 (0.025)	[+0.012, +0.039] (0.0070)	[+0.010, +0.043]	0.033 / 0.020
Linear-4D − LSTM+PI-QNN	−0.006	0.0069	−0.88 (0.376)	[−0.023, +0.008] (0.0082)	[−0.026, +0.013]	0.019 / 0.023
Linear-4D − LbL-QNN	−0.009	0.0065	−1.41 (0.158)	[−0.027, +0.005] (0.0084)	[−0.031, +0.009]	0.018 / 0.024
Linear-4D − LSTM+QTL	−0.003	0.0075	−0.34 (0.736)	[−0.018, +0.012] (0.0075)	[−0.020, +0.015]	0.021 / 0.021
LSTM − LSTM+PI-QNN	+0.008	0.0055	1.48 (0.139)	[−0.0005, +0.017] (0.0044)	[−0.005, +0.023]	0.015 / 0.012
LSTM − LbL-QNN	+0.005	0.0049	1.04 (0.299)	[−0.008, +0.018] (0.0067)	[−0.011, +0.021]	0.014 / 0.019
Extractor (= LSTM) − LSTM+QTL	+0.012	0.0044	2.69 (0.007)	[+0.003, +0.020] (0.0044)	[−0.002, +0.025]	0.012 / 0.012

Random Forest exceeds the hybrid by 0.020 AUC. The procedures disagree, as for RF − LSTM in Section 4.1: both bootstrap intervals exclude zero (year-block $[+0.007, +0.031]$; years×seeds $[+0.006, +0.034]$), whereas DeLong does not reject the null ($p = 0.074$; Holm-adjusted $p = 0.22$ across the three comparisons of this family), and the year-block standard error (0.0062) is again about half of the DeLong one (0.0109). The bootstrap intervals are pointwise and not multiplicity-adjusted. Under the alternative label definitions of Section 4.3 (Table 9) the difference stays between +0.011 and +0.020, with DeLong p from 0.074 to 0.367.

The hybrid is not distinguished from the circuit-free classical control: the control scores 0.009 below the hybrid, and the year-block and years×seeds intervals run from −0.026 to +0.006 and from −0.030 to +0.008, so the data are compatible with a small advantage of either model; no equivalence margin was specified in advance, and this is not an equivalence test. Against the standalone LSTM the difference is +0.005 with intervals that include zero. A bottleneck-free LSTM with the same hidden size, two further 4-dimensional controllers with other activations and full-feature XGBoost, evaluated in the canonical pipeline (Section S6.8), score 0.6435–0.6575. These experiments do not establish the four-dimensional bottleneck as the cause of the observed Random Forest–hybrid gap. The quantum parameters are 0.9 % of the trainable parameters of `LSTM+QNN`, so most trainable parameters are classical, although this ratio alone does not quantify the functional contribution of the circuit.

The three further hybrids behave like `LSTM+QNN` (Table 14). Random Forest exceeds them by 0.020–0.026, with year-block and years×seeds intervals that exclude zero in all three cases and DeLong p of 0.025–0.089; none is distinguished from the Linear-4D control (differences −0.003 to −0.009, all intervals include zero), and `LSTM+PI-QNN` and `LbL-QNN` are not distinguished from the standalone LSTM. `LSTM+QTL` trains a quantum head on a frozen pre-trained extractor, and in each seed the extractor is the standalone LSTM of that seed. The hybrid scores 0.012 below its own extractor (DeLong $p = 0.007$, year-block interval [+0.003, +0.020]); the years×seeds interval, which also resamples seeds, includes zero ([−0.002, +0.025]). In this experiment, then, replacing the classical head with the quantum head lowered the AUC rather than raising it. With RF − `LSTM+QTL` ($p = 0.025$) and RF − `LSTM+PI-QNN` ($p = 0.0497$), it is one of the three hybrid comparisons in which DeLong and the year-block interval both reject zero at the nominal level; these are secondary analyses and were not adjusted for multiplicity.

5 Discussion

5.1 What a Benchmark of This Size Can and Cannot Conclude

Under the iid DeLong approximation the estimated 80 %-power MDE for the three classical seed-mean comparisons on 528 correlated test months is 0.031–0.044 AUC; block-based estimates differ, particularly for RF − LSTM (0.017). The observed gaps between the architectures studied (0.003–0.03) are mostly smaller than these values. This does not mean that the models are equal. Three observations follow. First, for the individual-AUC scale comparison reported here, conditional test-sampling variability exceeded across-seed variability (seeds are 2–7 % of the combined variance of the classical models); this is a comparison of scales and not an estimate of how much of the total uncertainty additional seeds could remove. Second, for gaps of this size the conclusion depends on the estimand and on the procedure: single-seed verdicts depend on which seeds are compared (RF > XGBoost is declared significant in 27 % of the 15×15 seed pairings), and for the seed means DeLong and the bootstrap disagree on RF − LSTM and on RF − `LSTM+QNN`. Third, the reference points are strong: the observed mean RF–climatology AUC difference is about 0.003, no model improves the climatological Brier score on average (Random Forest matches it, the other models are worse), within a station no model discriminates better than the month×station climatology, and the reported comparisons do not establish skill beyond the seasonal-geographic base rate for any of the architectures evaluated.

For the quantum results, `LSTM+QNN` scores 0.020 AUC below Random Forest fitted on the same rows

(year-block interval +0.007 to +0.031, whereas DeLong does not reject the null, $p = 0.074$; across the label scenarios of Section 4.3 the difference stays between +0.011 and +0.020 with p from 0.074 to 0.367) and is not distinguished from a circuit-free classical control (the control scores 0.009 lower; intervals from about −0.03 to +0.008). The three further hybrids lie 0.020–0.026 below Random Forest and are not distinguished from the control either, and `LSTM+QTL` scores 0.012 below its own frozen classical extractor; on proper scoring all four hybrids are worse than the climatology and less well calibrated than Random Forest (Table 10). The available evidence does not establish a reproducible quantum performance advantage in this case study, and uncertainty remains substantial for differences of the magnitude observed here; the results neither show nor rule out an advantage on larger or richer data.

5.2 Recommended Reporting Protocol

Table 15 lists the practices that follow from the results. Items 1, 2 and 4 need only the test probabilities of the models; item 3 needs a specified and executed generating mechanism.

Table 15: Reporting protocol for small QML benchmarks.

#	Practice
1	State the estimand (single run or seed average; conditional on which sources) and report paired confidence intervals for it, with DeLong and a block bootstrap over the natural cluster (here calendar years); report discrepancies between procedures instead of selecting one.
2	Report the MDE as a measure of design sensitivity next to the intervals; do not treat it as a significance threshold or classify gaps below it as uninformative.
3	Check the procedures against a simulation with known differences at the sample size and dependence structure of the study.
4	Report across-seed variability separately from test-sampling uncertainty; if a relative-scale ratio is used, state its conditional and descriptive interpretation, and treat seed-level tests as statements about optimization, not generalization.
5	Include a climatological (or base-rate) reference with a proper scoring rule, and report skill relative to it.
6	Fit all model families on the same rows and tune them under a comparable budget, or state and quantify the asymmetry.
7	Add a classical control at the same low-dimensional interface as the quantum circuit.
8	Audit label validity (here, incomplete months) and re-evaluate the comparisons under alternative labels.
9	Choose equivalence margins by scientific relevance, independently of the MDE and of the results, include sampling error, and treat an interval wider than the margin as inconclusive.
10	When the target threshold is pooled across heterogeneous sites, report a site-only baseline and a within-site AUC, so that skill is not credited for separating sites with different base rates.

5.3 Limitations

Inferential procedures. The block bootstrap relies on 19 calendar years and, for RF − LSTM, gives about half the DeLong standard error; the iid estimator ignores temporal dependence. The simulations (Section 4.2) show how the procedures behave under stylized mechanisms (Gaussian binormal scores, shared and scorer-specific year effects, no spatial structure, no optimization noise): DeLong was anticonservative when year effects were scorer-specific, and the block bootstrap was somewhat liberal; the discrepancy observed for RF − LSTM (block standard error smaller than DeLong) is in the opposite direction to scenarios C–E, where the block standard error exceeded or tracked DeLong, whereas in the panel-matched check both estimators were closely aligned (0.011–0.012). With a small number of clusters ($K = 19$), a block standard error below the iid value can arise from negative within-year correlation of

the paired-difference components or from the downward bias and instability of cluster-bootstrap variance estimates with few clusters; it is not evidence of variance reduction. The simulations do not establish which procedure is best calibrated for the observational panel, so both are reported and their discrepancy is treated as inferential sensitivity. A delete-one-year jackknife (Section 3) sides with the block bootstrap: for RF − LSTM its standard error is 0.0062 (block 0.0061, DeLong 0.0112) and its t_{18} interval [+0.002, +0.028] excludes zero; for RF − XGBoost and XGBoost − LSTM it gives 0.0121 and 0.0152, and for the hybrid comparisons of Table 13 0.0062 (RF; interval [+0.007, +0.033]), 0.0087 (linear control) and 0.0043 (LSTM). Both cluster-based procedures rely on the same 19 years, so their agreement does not settle which estimate is calibrated for the panel. A circular moving-block bootstrap with blocks of 2, 3 and 4 consecutive years (`exp19_block_length.py`) changes the verdict of one comparison only: the RF − XGBoost interval, which includes zero with one- and two-year blocks, excludes it with three- and four-year blocks. The standard errors mostly decrease as blocks lengthen (RF − LSTM from 0.0063 to 0.0038; RF − XGBoost from 0.0119 to 0.0082), the opposite of what positive dependence between consecutive years would produce, and with 19 years a four-year block leaves only five independent draws, so these estimates are themselves unstable. The year is therefore kept as the cluster. 2026 is a partial year.

No retraining under alternative labels. The label scenarios re-evaluate fixed predictors; whether the Random Forest–hybrid gap would persist if the models were retrained under another label definition was not examined.

Hybrid models. The four variational hybrids have 15-seed archived predictions in the train-only pipeline; the quantum-kernel SVM and the encoding-bandwidth variants of the Supplement do not. The comparisons concern 4-qubit circuits in noiseless simulation, the analyses of the three hybrids other than `LSTM+QNN` were added after the main analysis, and no equivalence margin was specified in advance. An earlier execution of the core experiment used a dataset build in which two features (`et0_verano_prev`, `potencial_convectivo`) took, in about 68 % of rows, the value of another station on the same date, and in which P_{60} was recomputed as 28.00 mm (three training or validation labels differ); it gave RF − `LSTM+QNN` = +0.029 (p = 0.026) and Linear-4D − `LSTM+QNN` = +0.001. The corrected build reported here is the one of the classical rerun. The change of 0.009 in the first difference is smaller than its standard error, but it illustrates how much a seed-averaged hybrid comparison can move with a preprocessing detail.

Label validity. Incomplete months are 23 % of the test partition (Section 4.3). The models were not retrained under the alternative label definitions, months with incomplete coverage among the 12 antecedent months of a window were not examined, and 897 station-months with at least 10 observed precipitation days do not enter the panel because of the inner join and were not examined.

Baseline tuning and scope. Hyperparameters were set manually and no reproducible tuning history is available (Section S3); an Optuna-tuned Random Forest on 2,273 rows (0.6798 ± 0.0045) scores slightly below the untuned one on the same rows, so tuning moves the AUC by less than the noise, but XGBoost (0.662) should not be read as a ceiling for boosted trees. The panel is small (3,363 station-months), monthly aggregation does not resolve convective episodes, and the chronological split falls in a period of accelerated warming, so with one chronological split we cannot determine how distribution shift modifies model performance or sensitivity to optimization randomness. We evaluate one ansatz family in noiseless simulation with a 0.9–3.9 % quantum share of trainable parameters; inference-time gate noise changes the results little (Section S6.13) but says nothing about noise during training (Wang et al., 2021).

Generality beyond this dataset. All empirical conclusions concern the complete-variable panel retained by the processing pipeline, not the full set of station-months with precipitation observations (897 station-months with at least 10 observed days do not enter it; Section 4.3), and the evidence is empirical for one panel, but the sampling-error argument does not depend on the climate application. For a binary classifier with AUC A, n_1 positives and n_0 negatives, the Hanley–McNeil approximation (Hanley & McNeil, 1982) gives the standard error of one AUC from (A, n_1, n_0) alone; for the present test set it returns 0.024, matching the DeLong value. Assuming a correlation of 0.75 between the AUCs of two models scored on the same rows (the paired standard errors measured here imply 0.72–0.83), Table 16 gives the approximate 80 %-power MDE for other test-set sizes: about 0.049 for $N = 500$ at prevalence 0.41 and about 0.086 at prevalence 0.10. These values are specific to the assumed AUC (0.68), inter-model correlation and iid approximation; an MDE is a design sensitivity and not a bound on detectable effects, and temporal and spatial dependence can change it in either direction for paired comparisons (for RF − LSTM the block standard error here is smaller than the DeLong one). They do not replace validation on further datasets.

Table 16: Approximate MDE for the AUC difference of two models on the same test set (Hanley–McNeil standard error, AUC 0.68, inter-model correlation 0.75, $\alpha = 0.05$, power 0.80). Test-set size N and positive-class prevalence π.

N	MDE, $\pi = 0.41$	MDE, $\pi = 0.10$
250	0.069	0.122
500	0.049	0.086
1,000	0.035	0.061
2,500	0.022	0.038
5,000	0.015	0.027

6 Conclusions

On a benchmark of 528 correlated test station-months, conditional test-sampling variability exceeded across-seed variability on the scale of one AUC. Under the iid DeLong approximation the estimated 80 %-power MDE for the three classical seed-mean comparisons was 0.031–0.044 AUC, with block-based estimates differing, particularly for RF − LSTM. Conclusions about gaps of one to two hundredths depend on the inferential procedure: Random Forest against XGBoost is borderline, and for Random Forest against the LSTM the year-block bootstrap intervals exclude zero whereas DeLong does not reject the null. In the simulated scenarios, DeLong inference was approximately calibrated with shared year effects and became anticonservative as scorer-specific year effects increased; the block bootstrap was less sensitive to the simulated scenario but not nominal (type-I error 0.074–0.084, coverage 0.916–0.926). Neither procedure is selected a priori, and their discrepancy is reported. The reported comparisons do not establish improvement over the climatology, which, fitted on the 1938–2007 training and validation years, matches the 15-seed Random Forest (AUC 0.686 against 0.683); within stations the training-period climatology has the highest AUC of all models, no model improves its Brier score on average, and the circuit-free classical control scores 0.009 below the quantum hybrid (intervals from about −0.03 to +0.008), while Random Forest exceeds the hybrid by 0.020 (year-block interval excluding zero, DeLong $p = 0.074$), a difference whose sign is stable across the label definitions (DeLong p from 0.074 to 0.367). Three further hybrids give the same picture (0.020–0.026 below Random Forest, not distinguished from the control). For the classical models, the defensible conclusion is not that one of them is superior, but that this benchmark

cannot resolve differences of this size reliably; the evidence that Random Forest exceeds the hybrids is consistent in sign across hybrids and label definitions, but depends on the inferential procedure and does not survive multiplicity correction. About 23 % of the test months are incomplete. Re-evaluating the saved predictions of the three classical models on subsets of months and against alternative labels kept the sign of every paired difference but not its nominal significance: among the eight label scenarios the year-block interval of RF − LSTM excludes zero only under the original labels, and that exclusion is narrow and should not be interpreted as robust evidence, while RF − XGBoost reached $p = 0.019$ only with labels from the unjoined precipitation sum. No model was retrained under alternative labels.

The consequence for QML benchmarking is practical. Single-seed comparisons, and multi-seed comparisons that ignore sampling error, can report differences that the benchmark cannot reliably support, and the inferential target and procedure must be stated. Reporting the estimand, the noise budget with its MDE, a simulation check, a climatological reference and a label audit makes the evidential scope of a small benchmark explicit and separates claims the data can carry from claims they cannot.

Data and Code Availability

Code, processed panel data and model checkpoints are available at https://github.com/enriquerb32/limits-of-resolution-qml. The canonical pipeline is anchored in checkpoint `01_data.pt` (monthly panel N = 3,363, 1938-03 to 2026-05; windowed dataset N = 3,315; train/validation/test 2,273/514/528; 35 input features); thresholds (P_{60} = 28.1 mm, P_{90} = 86.32 mm) were computed on the training partition only. The daily station records are sourced from AEMET's public data holdings via *datosclima.es* under AEMET's Open Data reuse framework; only the processed, aggregated monthly panel is redistributed (`results/panel_checkpoint.csv`), with attribution to AEMET. Scripts for the analyses of this paper are organized under `code/local_experiments/` (`exp7_noise_budget.py`, `exp10_paired_ci.py`, `exp11_simulation_known_truth.py`, `exp12_monthly_coverage_audit.py`, `exp13_label_sensitivity.py`, `exp14_innerjoin_audit.py`, `exp15_panel_matched_simulation.py`, and `exp16_block_c.py` for the seed shares of differences, the year jackknife and the additional climatological references, computed from the archived probabilities without retraining and saved in `results/local_exp16_block_c.json`); figures are generated from saved results via `scripts/make_paper_figs.py`. The 15-seed QML core experiment (Table 12) is produced by `code/local_experiments/exp17_hybrids_multiseed.py` with the options `--fix-et0 --p60 28.1` and all seven models (outputs `results_hybrids_trainonly_fixet0_p60_28.1.json` and `results_hybrids_trainonly_fixet0_p60_28.1_probs.npz`); the earlier execution (`qml_core_trainonly.py`, `results_qml_core_trainonly_probs.npz`) is kept for reference. The paired comparisons of Tables 13 and 14 are produced by `exp10_paired_ci.py` with the option `--extra` pointing to the new archive, the label sensitivity of the hybrid (Table 9) by `exp13_label_sensitivity.py` with the same option, the seed shares and year jackknife by `exp16_block_c.py-{}-core` (`results/local_exp16_block_c_fixet0.json`), and the Random Forest refits by pipeline and fitting rows by `exp20_rf_fitting_rows.py` (`results/local_exp20_rf_fitting_rows.json`); test metadata are exported by `export_test_meta.py`. The commands, seeds and output files are listed in the repository README. Internal variable names (`dana_*`, `acp_*`) and the feature glossary are in the Supplement.

Declaration of Generative AI in Scientific Writing

During the preparation of this manuscript, the author used generative AI tooling for editorial language polishing, LaTeX formatting and structural drafting, and for drafting analysis scripts that the author ran and checked. No reported test statistic, empirical metric or benchmark result was generated by AI models; the numerical results come from the released scripts. The author assumes full scientific and editorial responsibility for the content.

Author Contributions

E.R. conceived the study, curated the data, developed the models, performed the statistical analysis, and wrote the manuscript.

Competing Interests

The author declares no competing financial or non-financial interests.

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References

Alexandersson, H. (1986). A homogeneity test applied to precipitation data. *International Journal of Climatology*, 6(6), 661–675. https://doi.org/10.1002/joc.3370060607

Allen, R. G., Pereira, L. S., Raes, D., & Smith, M. (1998). *Crop evapotranspiration—Guidelines for computing crop water requirements—FAO Irrigation and drainage paper 56*. FAO, Rome.

Bazgir, A., & Zhang, Y. (2024). QESM: A leap towards quantum-enhanced ML emulation framework for Earth and climate modeling. *arXiv preprint arXiv:2410.01551*. https://arxiv.org/abs/2410.01551

Bouthillier, X., Delaunay, P., Bronzi, M., Trofimov, A., Nichyporuk, B., Szeto, J., Sepah, N., Raff, E., Madan, K., Voleti, V., Ebrahimi Kahou, S., Michalski, V., Serdyuk, D., Arbel, T., Pal, C., Varoquaux, G., & Vincent, P. (2021). Accounting for variance in machine learning benchmarks. *Proceedings of Machine Learning and Systems*, 3, 747–769.

Bowles, J., Ahmed, S., & Schuld, M. (2024). Better than classical? The subtle art of benchmarking quantum machine learning models. *arXiv preprint arXiv:2403.07059*. https://arxiv.org/abs/2403.07059

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3. https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

da Silva, M. H. F., de Jesus, G. F., Nascimento, C. M. S., da Silva, V. L., & Cruz, C. S. (2026). Exploring quantum machine learning for weather forecasting. *Brazilian Journal of Physics*, 56, 22. https://doi.org/10.1007/s13538-025-01941-4

DeLong, E. R., DeLong, D. M., & Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3), 837–845. https://doi.org/10.2307/2531595

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7, 1–30. https://jmlr.org/papers/v7/demsar06a.html

Dietterich, T. G. (1998). Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, 10(7), 1895–1923. https://doi.org/10.1162/089976698300017197

Fellner, T., Kreplin, D. A., Tovey, S., & Holm, C. (2026). Quantum vs. classical: A comprehensive benchmark study for predicting time series with variational quantum machine learning. *Machine Learning: Science and Technology*, 7(1), 010501. https://doi.org/10.1088/2632-2153/ae365f

Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems (NeurIPS)*, 35, 507–520.

Hanley, J. A., & McNeil, B. J. (1982). The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 143(1), 29–36. https://doi.org/10.1148/radiology.143.1.7063747

Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., & Meger, D. (2018). Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 3207–3214. https://doi.org/10.1609/aaai.v32i1.12007

Huang, H.-Y., Broughton, M., Cotler, J., Chen, S., Li, J., Mohseni, M., Neven, H., Babbush, R., Kueng, R., Preskill, J., & McClean, J. R. (2021). Power of data in quantum machine learning. *Nature Communications*, 12, 2631. https://doi.org/10.1038/s41467-021-22539-9

Kumar, T. P., Raghavender, K. V., Veeranna, T., Srikanth, G. K., & Routray, A. (2026). Hybrid variational quantum-classical data assimilation for numerical weather prediction using Lorenz system benchmarks. *Scientific Reports*, 16, 31979. https://doi.org/10.1038/s41598-026-60719-z

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1217–1241. https://doi.org/10.1214/aos/1176347265

Marchisio, A., Ebrahim, A., Innan, N., Kashif, M., & Shafique, M. (2026). QLIF-CAST: Quantum Leaky-Integrate-and-Fire for time-series weather forecasting. *arXiv preprint arXiv:2605.18333*. https://arxiv.org/abs/2605.18333

Mari, A., Bromley, T. R., Izaac, J., Schuld, M., & Killoran, N. (2020). Transfer learning in hybrid classical-quantum neural networks. *Quantum*, 4, 340. https://doi.org/10.22331/q-2020-10-28-340

McClean, J. R., Boixo, S., Smelyanskiy, V. N., Babbush, R., & Neven, H. (2018). Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1), 4812. https://doi.org/10.1038/s41467-018-07090-4

Nieto, R., Gimeno, L., de la Torre, L., Ribera, P., Gallego, D., García-Herrera, R., García, J. A., Nuñez, M., Redaño, A., & Lorente, J. (2005). Climatological features of cutoff low systems in the Northern Hemisphere. *Journal of Climate*, 18(16), 3085–3103. https://doi.org/10.1175/JCLI3386.1

Obuchowski, N. A. (1997). Nonparametric analysis of clustered ROC curve data. *Biometrics*, 53(2), 567–578. https://doi.org/10.2307/2533958

Pastori, L., Grundner, A., Eyring, V., & Schwabe, M. (2026). Quantum neural networks for cloud cover parameterizations in climate models. *Machine Learning: Earth*, 2, 015008. https://doi.org/10.1088/3049-4753/ae4981

Peterson, T. C., Easterling, D. R., Karl, T. R., Groisman, P., Nicholls, N., Plummer, N., Torok, S., Auer, I., Boehm, R., Gullett, D., Vincent, L., Heino, R., Tuomenvirta, H., Mestre, O., Szentimrey, T., Salinger, J., Førland, E. J., Hanssen-Bauer, I., Alexandersson, H., Jones, P., & Parker, D. (1998). Homogeneity adjustments of in situ atmospheric climate data: A review. *International Journal of Climatology*, 18(13), 1493–1517. https://doi.org/10.1002/(SICI)1097-0088(19981115)18:13<1493::AID-JOC329>3.0.CO;2-T

Preskill, J. (2018). Quantum computing in the NISQ era and beyond. *Quantum*, 2, 79. https://doi.org/10.22331/q-2018-08-06-79

Romero, R., Ramis, C., Sumner, G. N., & Alonso, S. (1999). Daily precipitation patterns in the Spanish Mediterranean area: an objective classification. *International Journal of Climatology*, 19(1), 95–112. https://doi.org/10.1002/(SICI)1097-0088(199901)19:1<95::AID-JOC344>3.0.CO;2-9

Schuld, M., & Killoran, N. (2022). Is quantum advantage the right goal for quantum machine learning? *PRX Quantum*, 3, 030101. https://doi.org/10.1103/PRXQuantum.3.030101

Schuld, M., Sweke, R., & Meyer, J. J. (2021). Effect of data encoding on the expressive power of variational quantum machine learning models. *Physical Review A*, 103(3), 032430. https://doi.org/10.1103/PhysRevA.103.032430

Schwabe, M., Pastori, L., de Vega, I., Gentine, P., Iapichino, L., Lahtinen, V., Leib, M., Lorenz, J. M., & Eyring, V. (2025a). Opportunities and challenges of quantum computing for climate modelling. *Environmental Data Science*, 4, e35. https://doi.org/10.1017/eds.2025.10010

Schwabe, M., Pastori, L., Sarandrea, V., & Eyring, V. (2025b). Quantum machine learning for climate modelling. In *2025 IEEE International Conference on Quantum Artificial Intelligence (QAI)* (pp. 73–78). https://doi.org/10.1109/QAI63978.2025.00019

Skolik, A., McClean, J. R., Mohseni, M., van der Smagt, P., & Leib, M. (2021). Layer-by-layer training of quantum neural networks. *Quantum Machine Intelligence*, 3(1), 5. https://doi.org/10.1007/s42484-021-00047-9

Wang, S., Fontana, E., Cerezo, M., Sharma, K., Sone, A., Cincio, L., & Coles, P. J. (2021). Noise-induced barren plateaus in variational quantum algorithms. *Nature Communications*, 12, 6961. https://doi.org/10.1038/s41467-021-27045-6

A Computations

Estimators. For two models scored on the same 528 rows, the DeLong estimator gives the covariance of the two AUCs from their structural components (DeLong et al., 1988); it treats station-months as independent. The block bootstrap resamples the 19 calendar years (2008–2026) with replacement, keeping all stations of a year together, recomputes both AUCs on each resample ($B = 2{,}000$) and takes the standard deviation of the paired difference as its standard error (Künsch, 1989). With 19 clusters the estimate of the standard error is imprecise, and the direction of its discrepancy with the DeLong value depends also on the dependence structure (here the block standard error is below the DeLong value for RF − LSTM: 0.010 versus 0.014 for single seeds, 0.006 versus 0.011 for seed means). The two-way bootstrap draws a calendar-year resample and, independently for each model, a resample of its 15 seeds, and recomputes the difference of seed-averaged AUCs.

Seed-mean statistic. The AUC of one model is the mean of the DeLong structural components over positives and over negatives, so the mean of the 15 AUCs equals the AUC statistic built from components

averaged over seeds, and the paired DeLong variance of $\hat{\Delta}$ follows from the averaged components. This treats the 15 seeds as fixed and so conditions on the fitted models; the two-way bootstrap relaxes this by resampling seeds.

Seed share of variance. For model m with seed standard deviation s_m and sampling standard error SE_m, the share is $s_m^2/(s_m^2 + \mathrm{SE}_m^2)$. Using the mean DeLong SE over the 15 seeds, Random Forest (SD 0.0035, SE 0.0232), XGBoost (0.0054, 0.0236) and the LSTM (0.0063, 0.0237) give 2.3 %, 4.9 % and 6.6 % (2 %, 5 % and 7 % in Table 3). For the hybrids of the core experiment, `LSTM+QNN` (SD 0.0098, SE 0.0237), `LbL-QNN` (0.0129, 0.0238), `LSTM+QTL` (0.0147, 0.0239) and `LSTM+PI-QNN` (0.0170, 0.0238) give 14.7 %, 22.7 %, 27.4 % and 33.9 %.

Simulation mechanism. For each replication, years $u = 1, \ldots, 19$ receive a prevalence effect $\tau_\pi\, \eta_u$ on the logit of $\pi = 0.41$ and a number of rows of 528 split as equally as possible (or, in the unequal scenario, a multinomial split with Dirichlet(3) shares). Scorers $k \in \{A, B\}$ are $s_k = \mu_k y + e_k + \tau_s\, w_u + \tau_m\, v_{k,u}$, with $e_k = \sqrt{\rho}\, z + \sqrt{1-\rho}\, \varepsilon_k$ (so that the scorer noise has correlation ρ), w_u a year shift shared by both scorers and $v_{k,u}$ a year shift specific to each scorer; all $\eta_u, w_u, v_{k,u}, z, \varepsilon_k$ are standard normal and $\mu_k = \sqrt{2}\,\Phi^{-1}(\mathrm{AUC}_k)$ without year effects. The design AUCs without year effects are 0.65 for scorer B and $0.65 + \Delta$ for scorer A; the scorer noise has correlation $\rho = 0.9$, and the scorer-specific year shifts $v_{k,u}$ are independent between scorers. Scenario B has $\tau_\pi = \tau_s = 0.3$ and $\tau_m = 0$; C adds $\tau_m = 0.3$; D is C with Dirichlet year sizes; E has $\tau_\pi = \tau_s = \tau_m = 0.6$. The simulation target is the pooled population difference under the mechanism, with years drawn independently and weighted equally, that is, the difference for new years; in scenario D the shares of the 528 rows among years follow a Dirichlet(3) draw that is independent of the year effects, the sample statistic pools observations (so it weights years by their sizes), and the reference is the pooled population AUC with equally likely years of equal size. Scenario D changes the distribution of cluster sizes. Because the sizes are independent of the year effects, the pooled statistic is expected to be centered on the equal-weight reference; this was checked empirically and not proved: in D the mean of the estimates differs from the reference by +0.0002, −0.0008 and −0.0004 for design differences of 0, 0.015 and 0.03 (Monte Carlo standard error of the mean about 0.0008). The population AUC of each scorer is computed as the mean of four independent draws of 400,000 rows from the same mechanism with years drawn independently and equally weighted, and the Monte Carlo standard error of the population difference is reported. When the design difference is 0 the scorers are exchangeable and the population difference is set to 0. DeLong tests use the paired estimator of the seed-free case; the block bootstrap resamples years ($B = 300$ per replication).

Label re-evaluation (script `exp13_label_sensitivity.py`). Saved test predictions are linked to the monthly coverage table through the station and target month of each test window. Scenarios S0–S7 are the project labels on all rows; complete target months; target months with at least 25 valid days; rows with unambiguous labels (the plain and the proportional-coverage sums agree at the project threshold); proportional-coverage labels at the same threshold and at the recomputed threshold; and labels from the unjoined precipitation sum (script `exp14_innerjoin_audit.py`) at the project threshold and at the threshold recomputed from the unjoined training sums. S4–S7 evaluate predictors trained on the project labels against alternative labels; they do not evaluate a pipeline retrained under those labels.